Gram Schmidt orthonormalization

Let $A_1, A_2, \ldots, A_k \in \mathbb{R}^n$ be k-linearly independent vectors.

Let $Q_1 = \dfrac{A_1}{\|A_1\|}$

$$Q_2 = \frac{A_2 - \langle A_2, Q_1\rangle Q_1}{\| A_2 - \langle A_2, Q_1\rangle Q_1\|}$$

Observe that $Q_2^T Q_1 = 0 = Q_1^T Q_2$

Since $Q_1 = \dfrac{A_1}{\|A_1\|}$,

$$\text{Span}\{Q_1, Q_2\} = \text{Span}\{A_1, A_2\}$$

Assume that for $i < k$, we have got $Q_1, Q_2, \ldots, Q_i$ s.t

$$Q_r^T Q_s = Q_s^T Q_r = 0 \quad \text{for } r, s \le i$$

and

$$\text{Span}\{Q_1, \ldots, Q_i\} = \text{Span}\{A_1, \ldots, A_i\}$$

Let $Q_{i+1} = \dfrac{A_{i+1} - \left(\sum_{j=1}^{i} A_{i+1}^T Q_j\right) Q_j}{\| A_{i+1} - \left(\sum A_{i+1}^T Q_j\right) Q_j\|}$.

Then $Q_{i+1} \neq 0$.

For, if, $Q_{i+1} = 0$, then $A_{i+1} = \sum_{j=1}^{i} \left(A_{i+1}^T Q_j\right) Q_j$

$$\Rightarrow A_{i+1} = \text{Span}\{Q_1, \ldots, Q_i\}$$
$$= \text{Span}\{A_1, \ldots, A_i\}$$

This is a contradiction to the fact $\{A_1, \dots, A_k\}$ are linearly independent

Thus inductively, we get orthonormal vectors

$q_1, \dots, q_k$ s.t. for $1 \le i \le k$.

$$\text{Span}\{q_1, \dots, q_i\} = \text{Span}\{A_1, \dots, A_i\}$$

Notice that $$q_i = \frac{A_i - \left(\sum_{j=1}^{i-1} A_i^T q_j\right) q_j}{\| A_i - \left(\sum_{j=1}^{i-1} A_i^T q_j\right) q_j \|}$$

$$\Rightarrow A_i = \| A_i - \sum_{j=1}^{i-1} (A_i^T q_j) q_j \| q_i + \sum_{j=1}^{i-1} (A_i^T q_j) q_j$$

We write $$A_i = \sum_{j=1}^{i} r_{ji} q_j$$

Then $$A = [A_1, \dots, A_k]$$

$$= [q_1, \dots, q_k] \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1k} \\ 0 & r_{22} & \cdots & r_{2k} \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & r_{kk} \end{bmatrix}$$

$$= QR$$